IHP SG13G2 Shuttle — Density DRC Waiver Request

SVM5740: Support Vector Machine Inference Accelerator

Portland State University — Chip ID: SVM5740

Prepared for discussion with IHP shuttle coordinator — August 3, 2026

Summary: After an exhaustive fill campaign and three re-hardening attempts at varying placement densities, four local Metal2 density windows and one Metal3 density window in the SVM compute core cannot be brought into compliance with M2Fil.h / M3Fil.h local coverage rules. The violations are a physical consequence of the routing track density required by the design and cannot be resolved with fill insertion alone. We respectfully request a foundry waiver for these five windows.

Design Overview

Design name	SVM5740 (svm_compute_core + svm_top_ihp + chip_top)
Technology	IHP SG13G2, 130 nm SiGe BiCMOS
Chip area	2400 µm × 2400 µm
Core area	1400 µm × 1400 µm
Function	5-class SVM inference: 500 support vectors, 16-bit fixed-point
Flow	LibreLane 3.0.4 (OpenLane2-based), IHP PDK rev. c4b8b4e5
Routing layers	Metal1–Metal5 (RT_MAX_LAYER = Metal5)

Density DRC Rule Context

IHP SG13G2 density rules check both a global coverage ratio and a sliding-window local coverage ratio across the full chip:

Rule	Layer	Scope	Minimum
M2.j	Metal2	Global chip area	35%
M2Fil.h	Metal2 + Metal2:filler	Any 800 µm × 800 µm window	25%
M3Fil.h	Metal3 + Metal3:filler	Any 800 µm × 800 µm window	25%

The global Metal2 coverage **passes** at 35.72% (above the 35% M2.j threshold). Only the local sliding-window rules are violated.

Remaining DRC Violations

After the best-result fill run (minimum-size filler cells: 0.6 µm × 0.6 µm, minimum spacing: 0.42 µm), five local-window violations remain:

#	Rule	Window (µm)	Note
1	M2Fil.h	(400, 400) – (1200, 1200)	SW quadrant of compute core
2	M2Fil.h	(400, 800) – (1200, 1600)	W / centre of compute core
3	M2Fil.h	(800, 400) – (1600, 1200)	S / centre of compute core
4	M2Fil.h	(800, 800) – (1600, 1600)	Centre of compute core
5	M3Fil.h	(400, 400) – (1200, 1200)	Same SW quadrant (Metal3)

All five windows are located **entirely within the 1400 µm × 1400 µm compute core**, at chip coordinates 400–1600 µm in both X and Y. No violations exist outside the core (IO ring, sealring, pad areas are clean).

Physical Root Cause

Why the core is dense

The SVM inference engine computes a dot product between a 1024-bit input feature vector and each of 500 support vectors, accumulates five class scores, and returns an argmax — all in a single clock cycle at 40 MHz. The combinational logic requires:

- 500 × 64 multiply-accumulate units in parallel
- Dense interconnect on Metal2 (horizontal routing) and Metal3 (vertical routing) spanning the full core area
- Placement density of 45% (cells cover 45% of core floorplan area)

Why fill cannot reach 25% in these windows

Metal2 filler cells require a minimum clear gap between the filler edge and any existing signal wire. The relevant IHP design rules are:

Rule	Parameter	Value
MFil.b	Minimum filler cell width	0.6 µm
MFil.c	Minimum filler-to-signal spacing	0.42 µm
MFil.a2	Maximum filler cell width	5.0 µm

A single filler cell therefore requires a clear track of at least:

$$w_{\min} + 2 \times s_{\min} = 0.6\,\mu\text{m} + 2 \times 0.42\,\mu\text{m} = \mathbf{1.44}\,\mu\text{m}$$

In the four failing windows, the signal routing on Metal2 leaves no gap wider than ≈ 1.44 µm between adjacent tracks. This was confirmed empirically:

1. A targeted fill pass restricted *only* to the failing windows (job 117322, completed in 1m 45s) inserted **zero fill cells** — every candidate location was rejected by spacing DRC.
2. Three re-hardening attempts at progressively lower placement densities (45%, 42%, 40%) were run to loosen routing and create fill room. At each density the fill campaign was repeated with the smallest legal filler (0.6 µm cells). Results are summarised below.

Placement density	M2 global	M2Fil.h viol.	M3Fil.h viol.	Total
45% (original v1.0.0)	35.72%	4	1	**5**
42% (re-harden)	35.70%	4	2	**6**
40% (re-harden)	35.70%	4	2	**6**

Reducing placement density did not resolve the M2Fil.h violations: the routing tracks in those windows are spaced by timing-driven routing constraints, not by placement density alone. Furthermore, reducing density below 45% caused Metal3 local coverage to *decrease* (sparser routing lowers the Metal3 baseline), introducing additional M3Fil.h violations.

Conclusion

The five remaining violations represent a **hard physical limit**: the minimum legal filler cell cannot fit into the gaps left by signal routing in these windows. No fill insertion strategy — regardless of cell size, spacing tuning, or core re-hardening — can bring these windows into compliance without fundamentally reducing the number of support vectors or the core clock frequency.

Waiver Justification

1. **Global density is compliant.** Metal2 global coverage is 35.72%, above the 35% M2.j minimum. The density rule intent — ensuring uniform mechanical stress and CMP planarity across the chip — is satisfied globally.

2. **Violations are confined and small.** All five violations are within a 1200 µm × 1200 µm sub-region of the 2400 µm × 2400 µm chip. The IO ring, pad cells, power grid, sealring, and all areas outside the compute core are fully DRC-clean.

3. **The local under-density is modest.** Window Metal2 coverage is estimated at 22–24% in the failing windows (just below the 25% threshold), not a gross deviation.

4. **No adjacent large features.** There are no large metal plates (> 35 µm × 35 µm) that would trigger the Slt.i slotting rules. The routing tracks are narrow signal wires and narrow filler cells.

5. **Student research shuttle.** This is a first silicon submission for a graduate research project at Portland State University. The chip is not a production part; the density windows present no reliability risk for characterisation.

Request

We respectfully request a **foundry DRC waiver** for rules M2Fil.h and M3Fil.h in the five windows listed in Section 3, on the grounds that:

- Global Metal2 density is compliant (M2.j passes).
- The local under-density results from an unavoidable physical constraint (signal routing track pitch < minimum fill cell + 2×spacing).
- Extensive automated fill and three re-hardening iterations confirm no EDA-accessible remedy exists within the current design constraints.

Supporting artifacts — KLayout DRC report (`.lyrdb`), filled GDS, and LibreLane run logs for all three re-hardening attempts — are available on request.

Contact: Adam Handwerger, Portland State University `handwerg@pdx.edu`